GEK-22942A

INSTRUCTIONS

TACHOMETER MONITOR CARD

USED WITH

ADJUSTABLE SPEED DRIVES

SPEED VARIATOR PRODUCTS DEPARTMENT
ERIE, PENNSYLVANIA

GENERAL ELECTRIC

LIST OF EFFECTIVE PAGES

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

1.0 Introduction

This instruction provides basic information for the operation, adjustment and troubleshooting of the Tachometer Monitor Card(s). The only difference between the G02 and G03 cards is the G02 card provides a fault relay (FR) for use in drives that do not contain such a relay. This relay provides a means of shutting down the drive when a fault develops.

CAUTION

THIS CARD MAY BE WIRED TO PROVIDE AN INTERLOCK FUNCTION DEPENDING ON THE CONFIGURATION OF THE INTERCONNECTION WIRING TO THE CARD; REMOVAL OF THE CARD (WHEN OPERATING) MAY CAUSE THE DRIVE TO SHUT DOWN AND INHIBIT RESTART. IF AUTOMATIC SHUTDOWN IS NOT PROVIDED, (ON REMOVAL OF THE CARD) NO OVERSPEED OR LOSS OF TACHOMETER SIGNAL PROTECTION WILL BE PRESENT WITH THE CARD REMOVED.

1.1 Tachometer Monitor Card Purpose

To provide drive system protection against motor overspeed or loss of tachometer feedback signal by shutting the drive system down prior to equipment damage. Also to provide a visual indication that the card circuits have provided protection and have shut down the drive system.

1.2 Card Specifications*

Temperature Range (Ambient)
Normal Operating: 10C to 40C (50F to 104F)

Operating Voltage Requirements: See Table I

Normal Operating Current Required:

ac/dc tach	.7	ma max.
+20vdc supply	100	ma max. G02
+20vdc supply	42	ma max. G03
−20vdc supply	12	ma max.

Environmental Requirements:

This equipment may be stored at ambient temperature of −20C to 70C for a period of up to one year. Air must be free of chemical and electrically conductive contaminants, and other conditions must be such that no moisture condensation occurs in or on the equipment.

All relays or high transient devices must have suppressed networks.

In addition, when a control that has been in operation will be shut down for either a short or extended period of time, it is recommended the environmental conditions be maintained the same as when in operation. It is recommended that space heaters or equivalent devices be used to maintain the equipment in its normal operating environment (Temperature).

1.2.1 Replacement and Interchangeability

A G02 can replace a G03, but a G03 is not interchangeable with a G02 card.

NOTE

FIELD REPLACEMENT OF CARD COMPONENTS SHOULD NOT BE ATTEMPTED.

* These specifications are for the printed circuit card only; not system specifications.

1.3 Card Input-Output Requirements

The following table lists the function, voltage range and card/receptacle tab number. All measurements are with respect to common, (tab 23) except tachometer input.

FUNCTION	VOLTAGE RANGE	TAB	REMARKS
+DC Supply Input	20 VDC ± .1 Volt	26	
–DC Supply Input	20 VDC ± .1 Volt	13	
Armature Fdbk. Input	0 to 10 VDC Max.	8	Approx. 2.5V causes trip.
Tachometer Voltage Input (trip speed range)	55–87 VDC	(+)(–) 24–30	Tabs 28–29 25–27 jumpered
Tach Input Connection depends on volts per 1000 rpm tach output at motor top speed	82–145 VDC	(+)(–) 24–30	No jumpers
	120–210 VDC	(+)(–) 25–28	Tabs 29–30 24–27 jumpered
	190–360 VDC	(+)(–) 25–28	No jumpers
FR Coil Voltage Input (G02 only)	20 VDC ± .1 Volt	14	Tab 4–19 jumpered
Card Interlock Input (if used)	20 VDC or 115 VAC 3 amp cont. 6a Inrush Max.	2–3	Shuts system down when card is removed.
Zero Align (for loss of tach signal)	+.1 VDC to +.9V (set spd) 0 VDC to + 7.5V (changing speed)	9	
Trip Voltage Output	Approx. 2.0 VDC	4	
Non-trip Voltage Output	Approx. 12.0 VDC	4	
Reset Voltage Input	"Common"	4	
Auxiliary Input	Approx. 10.0 VDC	9	Causes trip
FR Relay (G02 only)	2 NO, 2 NC Contacts	See Diagram	

TABLE I

1.4 Card Adjustments

There are two adjustments provided on the G02 or G03 Tachometer Monitor card. The first potentiometer adjustment is "Zero-Align" which provides a means of balancing (zeroing) the output of the "loss of tachometer" circuit inputs (armature feedback voltage versus tach voltage) from 0.1V to +.5VDC. To check (adjust) the "Zero-Align" circuit, perform the following steps:

1.4.1 Zero Align Adjustment

a. Set motor speed control at the point that provides 30% to 50% of armature voltage.

b. Using a sensitive multimeter, verify that +.1V to +.9VDC (set speed-constant load) is measured between tabs 9 and 23 (common). If more than .9VDC is measured, slowly adjust the "Zero Align" potentiometer in such a direction to bring the reading within the required range.

c. Using the system "speed control" verify that tab 9 does not exceed +7.5VDC throughout the speed range of the motor [if reversing drive, verify in both direction modes and balance (zero) as required between "forward" and "reverse" to meet.

d. Set the system "speed control" to zero and shut off power to the system.

e. Disconnect the tach input leads (24 and 25) at the terminal board end (regulator terminal board) not at the card receptacle end. Make sure wire ends removed are not making contact with metal.

CAUTION

CHECK TO BE SURE THAT THE TACH SIGNAL, IF USED FOR SPEED REGULATOR, IS NOT DISCONNECTED BETWEEN THE REGULATOR AND THE TACH.

f. Apply power to the system and slowly increase the "speed control". Drive will "trip" (shut off) and "Fault" indicator will illuminate.

g. Return "speed control" to zero and press and release "reset" pushbutton; "Fault" indicator extinguishes.

h. Shut off power to the system and reconnect wire removed in Step e.

i. Apply power to drive system and check for normal operation.

The second card adjustment is the "Trip Speed" potentiometer which provides a means of setting the desired trip point for motor overspeed. To check (adjust) the "Trip "Speed" circuit, perform the following steps:

1.4.2 Trip Speed Adjustment

NOTE

THIS ADJUSTMENT MUST BE MADE WITH THE MOTOR IN THE "COLD" STATE.

a. Apply power to the drive system and slowly increase the "Speed Control" until motor reaches rated top speed. If drive trips prior to reaching motor rated top speed, turn the "Trip Speed Adj." potentiometer clockwise (CW) five turns, reset the system and increase "Speed Control" until rated top speed is reached.

b. To check trip, with the drive motor operating at rated top speed, counting the number of turns, slowly adjust the "Trip Speed" potentiometer counterclockwise (CCW) until the drive system shuts off.

c. Readjust the "Trip Speed" potentiometer clockwise (CW) to the original setting in Step b (to its former setting).

d. Restart the drive and using a hand tachometer, adjust the "Trip Speed" potentiometer for 200 rpm above motor rated top speed.

e. Sharply accelerate drive up to motor rated top speed and no trip should occur.

2.0 Principles of Operation

Section 2 will provide a brief theory of operation and troubleshooting information for the tachometer monitor card.

2.1 Theory of Operation

The tachometer monitor card consists of three operational amplifiers, two programmable unijunction transistors (PUT), an output SCR and associated components. As previously stated, the G02 card also contains a relay. Refer to Functional Block Diagram – Page 5.

2.1.1 Trip Speed Circuit

The trip speed circuit consists of a series resistor input network (for different tach voltage input ranges), a differential amplifier (to handle plus or minus voltage inputs), a PUT circuit with potentiometer to set the point of overspeed trip, and an output SCR with fault indicator which is utilized by both circuits on the card. Drives using d—c tachometers or reversing drives with d—c tachometers require a diode rectifier bridge with filter prior to the card input.

The tach signal is brought in through the correct resistance network to the differential amplifier which provides a positive output voltage. A positive voltage is required at the programmable unijunction transistor (PUT) anode to permit conduction. The PUT gate signal sets the point at which conduction will occur and is adjusted by the "Trip Speed" potentiometer. When the output of the differential amplifier exceeds the conduction point voltage as set by "Trip Speed" potentiometer, the PUT conducts which causes the output SCR to conduct, which in turn provides a decreasing voltage to the coil of an energized relay (picked up) to de-energize (drop out) causing the drive to shut down. The conduction of the SCR also causes the fault indicator to illuminate. The fault indicator will continue to be illuminated until a "common" potential (reset) is momentarily applied to Tab 4, which removes the conduction holding current to the SCR and causes it to return to its blocking state which will cause the fault indicator to extinguish.

2.1.2 Loss of Tachometer Signal Card

The loss of tachometer signal circuit consists of a series resistor input network (for different loss of tachometer percentage input shutdown levels) for armature feedback voltage. This armature voltage feedback signal must be scaled down to a 0 to 10 volts d—c maximum to be compared against the tachometer input signal. The armature feedback signal is processed by an absolute amplifier and associated circuitry, and in turn is used as an input to a second differential amplifier. The second input to this differential amplifier is the output of the first differential amplifier that was used for the "Trip Speed" circuit. This input from the "Trip Speed" differential amplifier is brought through the "Zero Align" potentiometer and into the other input of the second differential amplifier. The zero align potentiometer provides a means of nulling out (balancing) the difference to +.1 to .+9VDC at a set speed between the armature feedback voltage and the tachometer feedback voltage. Under normal operation, the comparison of the armature voltage feedback signal and the tachometer signal will cancel each other out in the 2nd differential amplifier resulting in a zero voltage output from the 2nd differential amplifier. In the event of the loss of tachometer signal, the differential amplifier will sense a difference when the tachometer voltage drops 25% or more as compared to the armature feedback voltage, which in turn will cause a positive voltage to be applied to a second PUT anode circuit causing it to conduct, which will initiate system shutdown through the SCR circuit as described in 2.1.1, thus providing a "loss of tach" trip. Wrong polarity tachometer connection will cause a similar trip unless rectified external to this card. The fault indicator and reset function described in 2.1.1 are also in this part of the circuitry.

2.2 Troubleshooting

Troubleshooting procedures required for this card are based on the input/output votlage requirements as listed in Table I. If the output requirements of the card fail to meet the requirements of Table I when all the input requirements have been verified as correct, the Card should be replaced. In the event a new card is placed in the drive system, follow the adjustment and checkout procedures listed under paragraph 1.4.

2.2.1 General System Symptoms

a. Drive system shuts down at very low speed.

1. Check polarity connection of tachometer input.
2. Check for presense of tachometer input.

b. Drive trips for overspeed conditions.

1. Check for loss of motor field.
2. Check for faulty tachometer connection or loss of tachometer.
3. Check system for motor speed overshoot due to drastic load changes or when accelerating to high safe speed. (Lower or lessen system response.)

2.2.2 Inputs

Verify that all inputs (observing polarity requirements) are as specified in Table I, paragraph 1.3.

2.2.3 Outputs

Verify that all outputs are as specified in Table I, paragraph 1.3.

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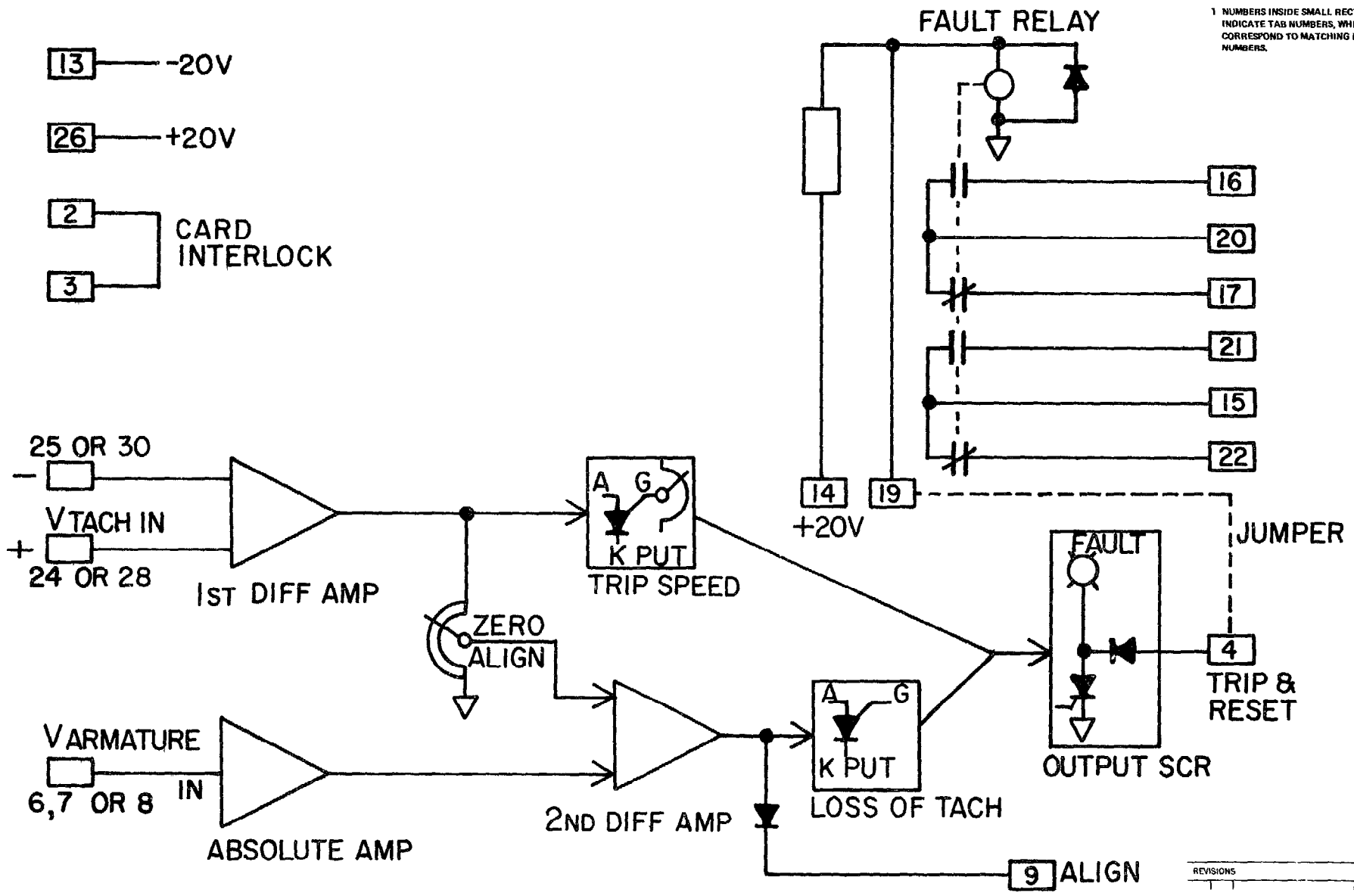
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UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE
FUNCTIONAL DIAGRAM
TACHOMETER MONITOR

FIRST MADE FOR STANDARD LINE
193X242AB G02

NOTES
1 NUMBERS INSIDE SMALL RECTANGLE INDICATE TAB NUMBERS, WHICH CORRESPOND TO MATCHING RECEPTACLE NUMBERS.



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